

Question 1: Forces and flight

[10 points]

An aircraft is flying straight and level at constant speed.

- (i) Name the four fundamental forces acting on the aircraft and state the direction of each force.
- (ii) What relationships must approximately hold between these forces during straight and level flight?
- (iii) Suppose the pilot increases thrust while the other conditions initially remain unchanged. Explain what you expect the aircraft to do.

Hint: Think about whether the forces are balanced or unbalanced.

Question 2: Using the lift equation

[10 points]

The lift generated by a wing is

$$L = \frac{1}{2} \rho V^2 S C_L$$

Consider an aircraft flying with

$$\rho = 1.2 \text{ kg/m}^3, \quad V = 20 \text{ m/s}, \quad S = 10 \text{ m}^2, \quad C_L = 0.5$$

- (i) Calculate the lift produced by the aircraft.
- (ii) If the aircraft speed increases from 20 m/s to 40 m/s while all other quantities remain unchanged, by what factor does the lift change?
- (iii) Explain why aircraft speed has such a strong effect on lift.

Question 3: Angle of attack and stall

[10 points]

Consider an aircraft whose pilot gradually increases the angle of attack.

Choose from: lift increases, flow separation, stall, lift decreases.

Before the stall angle is reached, increasing the angle of attack generally causes the _____.

Beyond a sufficiently large angle of attack, significant _____ occurs over the wing, which can lead to _____.

- (i) Fill in the blanks.
- (ii) Explain physically why increasing the angle of attack does not always increase lift.
- (iii) A student says: "To obtain more lift, the pilot can simply keep increasing the angle of attack." Is this statement true or false? Explain your answer.

Question 4: Wing geometry and induced drag

[10 points]

Two aircraft have wings with the same area $S = 20 \text{ m}^2$.

Wing A: span $b = 10 \text{ m}$

Wing B: span $b = 5 \text{ m}$

The aspect ratio is

$$AR = b^2 / S$$

- (i) Calculate the aspect ratio of Wing A and Wing B.
- (ii) Which wing has the higher aspect ratio?
- (iii) Which wing would generally produce less lift-induced drag under comparable conditions? Explain why.
- (iv) Briefly explain how wing-tip vortices are formed.

Hint: Think about the pressure difference between the upper and lower surfaces of a lifting wing.

Question 5: Aircraft static stability

[10 points]

An aircraft is initially flying in a trimmed condition. A gust suddenly causes its nose to pitch upward.

Aircraft A: develops a tendency to pitch back toward its original condition.

Aircraft B: remains at the new disturbed condition.

Aircraft C: develops a tendency to pitch even farther away from its original condition.

Choose from: statically stable, neutrally stable, statically unstable.

- (i) Classify Aircraft A, B, and C.
- (ii) What is meant by a restoring moment?
- (iii) For longitudinal static stability, after a small nose-up disturbance, should the resulting aerodynamic pitching moment tend to pitch the nose up or down? Explain.
- (iv) Why is static stability an important consideration when designing an aircraft?